



Documentation of my Final Assignment Process

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Initial Idea

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Einstellungsnr	Szenennr	Shotnr.	Location	Einstellungsgröße	Kamerabeweg.	Set - Blickwinkel/Fokuspunkt	Ideale Drehzeit	Beschreibung/Action	Besondere Ausrüstung	DigiBild	Bildschirm 1	Bildschirm 2	Tischbildschirm	Weiteres Extrafootage	Audio?
2	E1 (Bildschirm)	01	01A	LAT Frankfurt	CU	Push In	Kats Zimmer - Schreibtisch	Nacht	Wir sehen Kat am Schreibtisch sitzen und essen, Establishing Kat & ihr Zimmer		6	5	4	10	1, 4, 2	
3	E2 (Hinter Schreibtisch)	02	02A	INT Zimmer 1	MCU	Tilt	Kats Zimmer - (Richtung) Bett	Nacht	Essen		6	5	4	10		
4	E3 (Rahmen stehend)	02	02B	INT Zimmer 1	MCU	Static	Kats Zimmer - Schreibtisch (diagonal)	Nacht	Digitaler Bilderrahmen, Bild mit Ron erscheint		6	5	4	10		
5	E4 (Gegenshot Schreibtisch)	02	02D	INT Zimmer 1	MCU	Static/Tiny Push In	Kats Zimmer - Stuhl (diagonal)	Nacht	Lächeln & Löffel		6	5	4	10		
6	E2.2 (Rahmen Hand over the shoulder)	02	02C	INT Zimmer 1	MED	Push In (Slow)	Kats Zimmer - Schreibtisch (diagonal)	Nacht	Kat nimmt Bilderrahmen in die Hand (over the shoulder)		6	5	4	10		
7	E2.3 (Rahmen Hand)	02	02C	INT Zimmer 1	MCU	Static	Kats Zimmer - Schreibtisch (diagonal)	Nacht	Kat berührt Ron auf dem Bildschirm -> Animation + nächstes Bild		7	5	4	10		
8	E4	02	02D	INT Zimmer 1	MCU	Static	Kats Zimmer - Stuhl (diagonal) -> Rückseite vom Pad sichtbar	Nacht	Gegenshots Kats Reaktionen? + Sehnsüchtiger Ausdruck; ein Gedanke reißt sie aus den Erinnerungen, Konversation mit KI und Reaktion			5	4	10		
9	E5	02	02E	INT Zimmer 1	MED	Static	Kats Zimmer - Schreibtisch	Nacht	Chat öffnen							
10	E6 (Sellich)	02	02F	INT Zimmer 1	MED	Push In	Kats Zimmer - Schreibtisch (Sellich)	Nacht	Kats Finger schweben über dem Pad, tippen							
11	E4	02	02D	INT Zimmer 1	MCU	Static	Kats Zimmer - Schreibtisch (Sellich)	Nacht	Gesichtsausdruck bei fehlender Antwort							
12	E3	02	02B	INT Zimmer 1	MED	Static	Kats Zimmer - Richtung Bett	Nacht	Kat will weiter essen							
13	E5	02	02E	INT Zimmer 1	MCU	Static	Kats Zimmer - Schreibtisch (Diagonal)	Nacht	Antwort, Kat will Audio abspielen, Unterbrechung durch KI							
14	E7	02	02G	INT Zimmer 1	MCU	Tilt Up	Kats Zimmer - KI-Bildschirm	Nacht	KI-Bildschirm							
15	E4	02	02D	INT Zimmer 1	MCU	Static	Kats Zimmer - Richtung Bett	Nacht	Reaktion, Kat hört Nachricht an							
16	E2	02	02H	INT Zimmer 1	WS	Push Out (Tiny)	Kats Zimmer - Schreibtisch	Nacht	Kat hört Nachricht an und fühlt sich einsam							
17	E3	02	02B	INT Zimmer 1	MED	Static	Kats Zimmer - Schreibtisch (Sellich)	Nacht	Kat startet eigene Nachricht, bricht sie wieder ab							
18	E6	02	02F	INT Zimmer 1	CU	Static	Kats Zimmer - Schreibtisch	Nacht	Close-Ups Finger über dem Pad?							
19	E1	02	02E	INT Zimmer 1	MCU	Static	Kats Zimmer - Schreibtisch	Nacht	Schreibkonversation							
20	E3	02	02B	INT Zimmer 1	MED	Static		Nacht	Schreibkonversation							
21	E4	02	02D	INT Zimmer 1	MCU	Static		Nacht	Schreibkonversation							

SCENE NUMBERS

- 1. SCENE 01
- 2. SCENE 02
- 3. SCENE 03
 - 03A
 - 03B
 - 03C
- 4. SCENE 04

SCENE 03A

SEARCH

LOCATION | DAYTIME | SHOT POV | CHARACTERS | DESCRIPTION

Digi Lab Evening MCU +10 people fun, relaxed

NOTES

Mario Kart Event

turns into

MOODBOARD

CREATE PDF

Initial Goals

Office on the web Frame

- Improve readability
- Make it more accessible / more visually engaging
- Good UX/UI Design
- Incorporate dropdown menus instead of only tables

Office on the web Frame

- Streamlit (Design)
- CSV (File saving)
- Pillow (Pictures)
- Panda / Great Tables (Tables/Data Structure)

Select Data Time and Detector

streamlitapp want to find data?

By event name

Select Event

GW150914

Detector

H1

Gravitational Wave Quickview

- Use the menu at left to select data and set plot parameters
- Your plots will appear below

GW150914

GPS: 1152399402.4

Mass 1: 34.6 $M_{\odot}$

Mass 2: 30.0 $M_{\odot}$

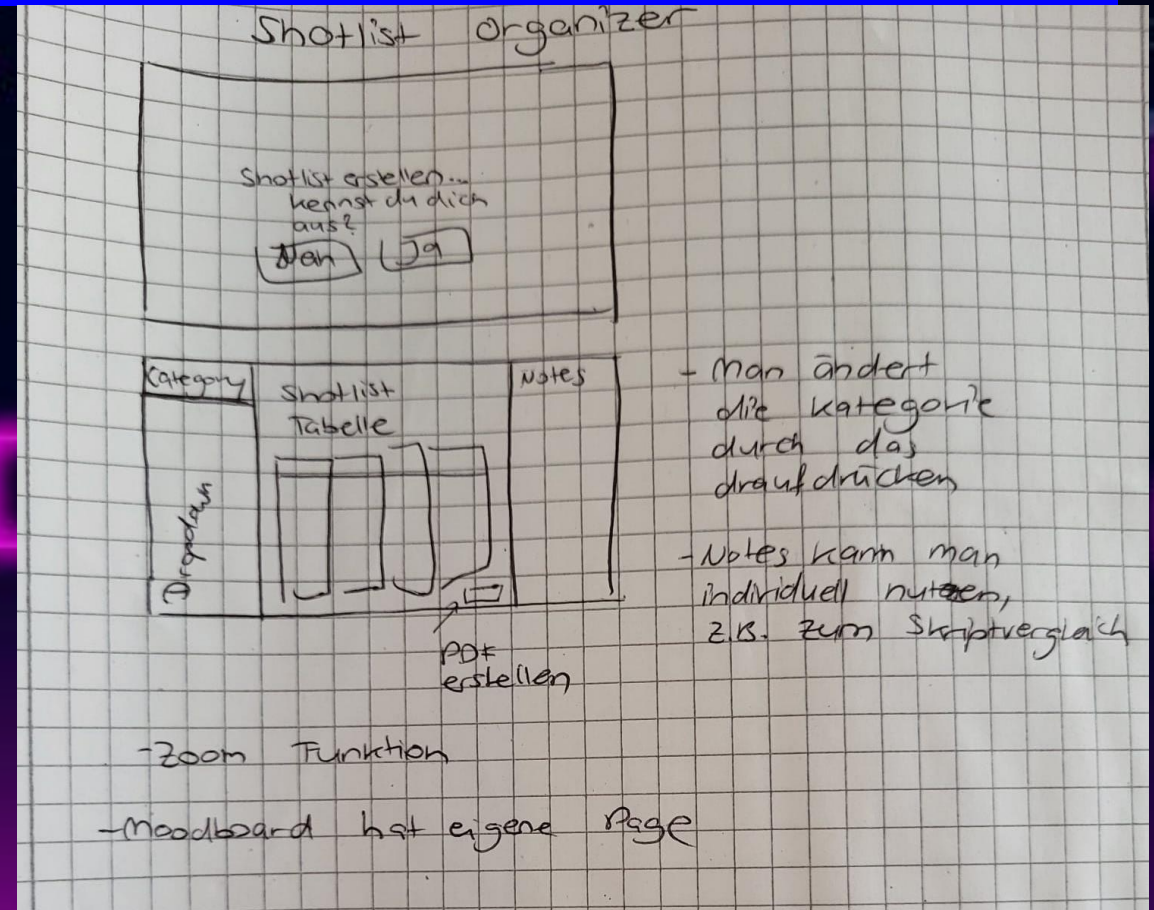
1. Create a visually engaging interface idea
2. Shotlist classes & Sort by (Dropdown Menu)
3. Details of the different interfaces
 - > sort by scene number
 - > sort by location
 - > sort by daytime
4. Moodboard
5. "Create PDF" button



I opened PyCharm and realized I had no idea
where to start...

So I needed to figure out the precise structure I wanted for my website:

This is how I noted down the original concept I had in mind



Figuring out the organization of the pages

I figured out what categories I wanted the shotlist organizer to display and started brainstorming

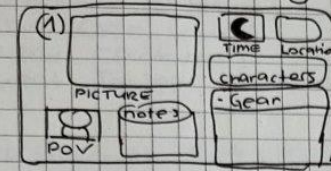
But I realized I was lacking a lot of understanding of what else I could and could not do with streamlit

Tech Basics

31.08.2026

idea:

- something to organize a shotlist



how to make it more python focused?

- schedule / time calculator
- save data & add data (like a notes app!) / logging in
- make it a bit fun / customize it

- maybe gamify it?

this is gonna be cute pls! :3

cozy <3 HTML elements at the end maybe :P (not the focus)

details:

perspective

vorgegebene auswahl
dropdown menu

- ↳ MC4 (medium Close Up)
- ↳ CU (Close Up)
- ↳ FS (Full Shot)
- ↳ MS (medium Shot)
- ↳ WS (Wide Shot)
- ↳ Other...

OTS (over the shoulder)

...type here...

adds it to the list of options (append)

notes

→ scene video :)

characters:

how many?

big group

other...

relationship?

lovers

family

• a pair

• one person

• friends

• enemies

• strangers

• other (type...)

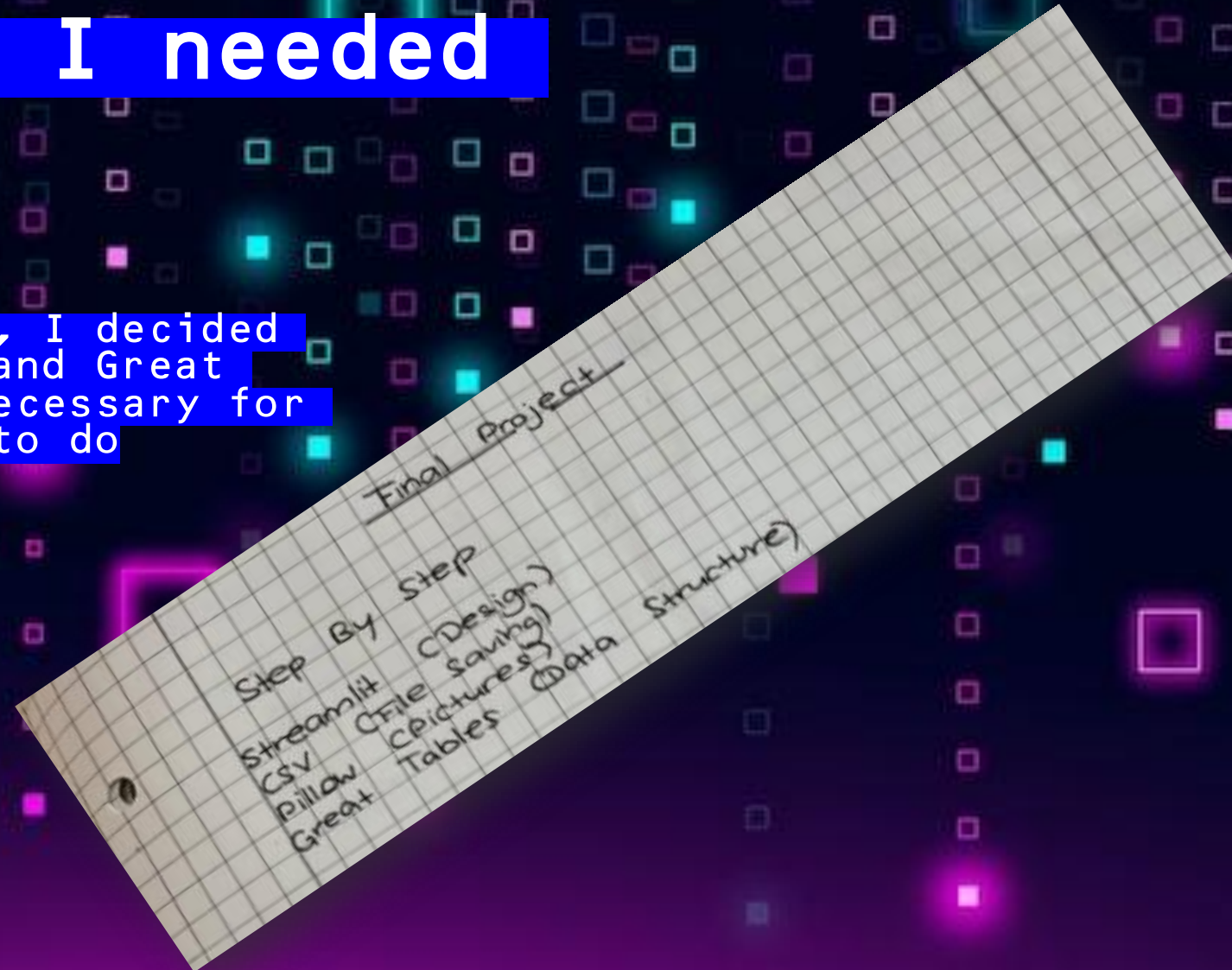
type in time suspected for this scene

location

- add manually

So then I needed to
structure what I needed
to learn...

Through this process, I decided
that both Pillows and Great
Tables would not be necessary for
what I wanted to do



Figuring out streamlit

I decided to separately practice the many different features of streamlit because I wanted to get to the point where I would grasp it so fundamentally that it would be easy to use

★★ Streamlit ★★★★★★
→ developing Web Applications

```
st.title ("Title")  
x = st.text_input ("What is the name of  
your project?")  
st.write (x)
```

→ docs. streamlit.io
→ Styling Stuff

• You can use Markdown Stylistics in strings!
They work :P

Working with data (CSV files)

```
import pandas as pd  
data = pd.read_csv ("movies.csv")  
st.write (data)
```

CSV interlude ☹☹☹☹☹☹☹☹

how to import?
just copy the path :)

♥ Streamlit ♥♥♥♥♥♥♥♥

multipage applications

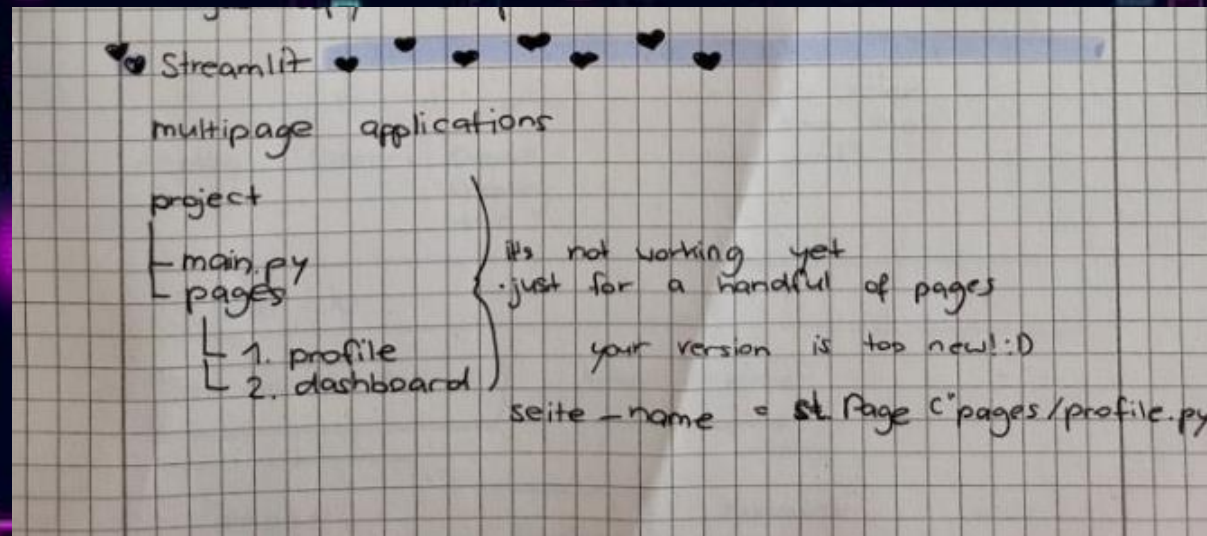
```
project  
├ main.py  
└ pages
```

```
├ 1. profile  
└ 2. dashboard
```

} it's not working yet
- just for a handful of pages
your version is too new! :D

seite - name = st.Page ("pages/profile.py"

Figuring out streamlit



Along the way I encountered problems such as: My newer version of streamlit at first not displaying and linking files together in the sidebar like it had been explained in tutorials

Figuring out streamlit

Streamlit

`st.menu-button` behaves like `st.button` except it returns multiple options instead of just `True`

`options (Iterable)`: list, set, anything supported by `st.dataframe`

`key`: optional string / integer to use as a unique key
`st.session_state[key]`

`type`: specifies the button type

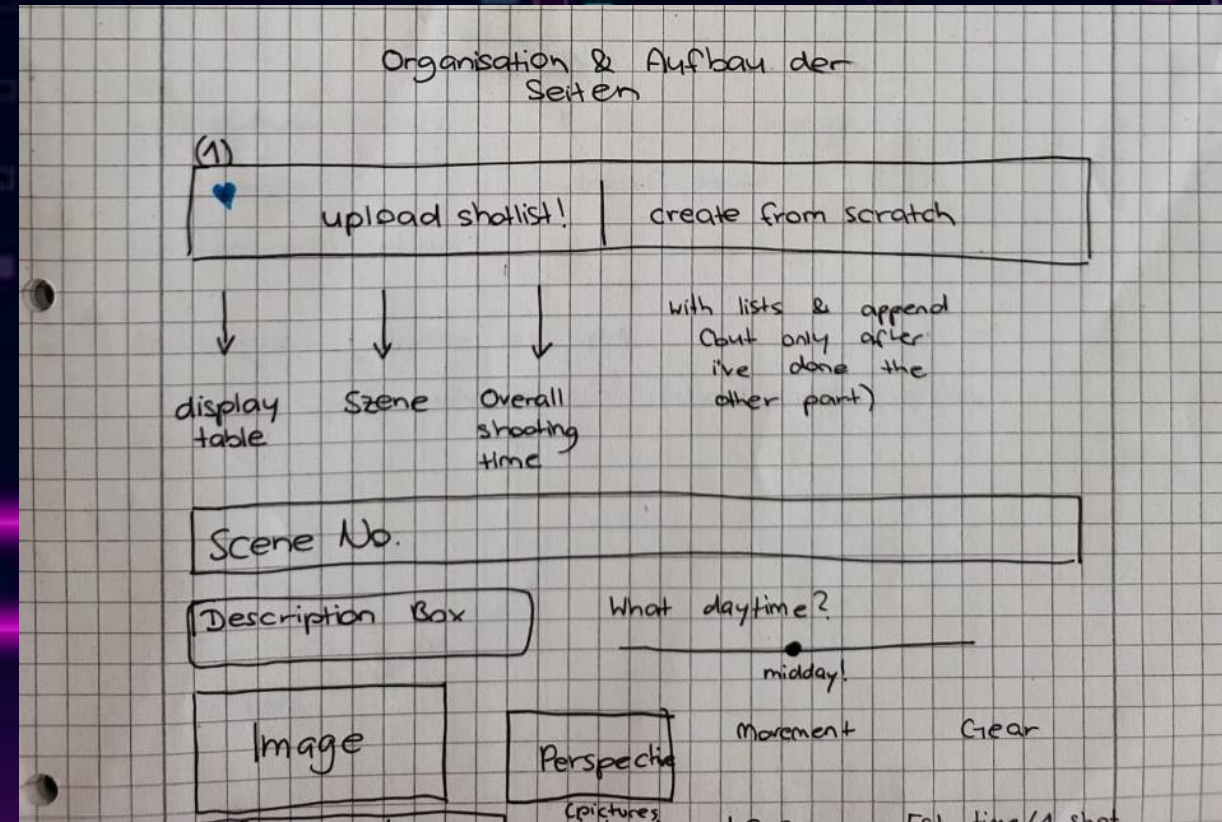
- `"primary"`: background is in the app's primary colour for emphasis
- `"secondary"`: the button's background does normal emphasis
- `"tertiary"`: subtle

`st_color_picker`: Displays a color picker widget

More and more my
understanding of
streamlit increased

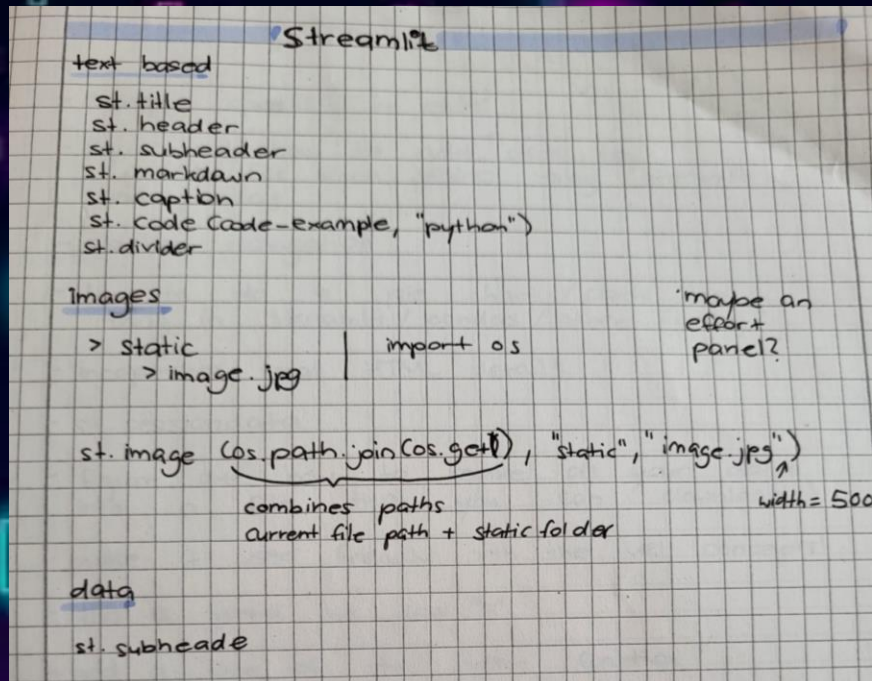
Figuring out streamlit

Which helped me develop a
more precise concept and
understanding of what I
wanted



Figuring out streamlit

But generally understanding it did not always translate into me understanding how I would have to specifically adjust it to my code. You can see the difference between my learning and final code



```
def image_creation():  
    image = st.text_input("Paste the URL to your image")  
    if image:  
        st.image(image)  
        st.badge(label="Good job!!", color="green")  
    else:  
        st.badge(label="No image yet!! Add one above 🐙", color="red")
```

Figuring out CSV

After this, I realized I was faster when trying things out in practice

CSV File Handling

read | write | transform

csv.writer

header

```
rows = [
    ['Shot Number', 'Calculated Time'],
    ['01', 'x + y'],
    ['02', 'x + y']
]
```

a list of lists :D

write mode

```
import csv
with open('results.csv', 'w', newline='') as file:
    writer = csv.writer(file)
    writer.writerow(rows)
```

or

if you write each individually (good for imports)

```
with open('results.csv', 'w', newline='') as file:
    writer = csv.writer(file)
    for row in rows:
        # perform an action needed for each row
        writer.writerow(row)
```

csv.DictWriter

list of dictionaries → (good for outside data :D)

```
fieldnames = ['name', 'score']
rows = [
    {'name': 'Alice', 'score': 95},
    {'name': 'Bob', 'score': 82}
]
```

```
with open('results.csv', 'w', newline='') as file:
    writer = csv.DictWriter(file, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(rows)
```

reader.fieldnames

```
reader = csv.DictReader(file)
new_fields = [ f for f in reader.fieldnames
               if f != 'email' + ['status'] ]
```

Figuring out Pandas

With Pandas I
practiced displaying
and changing bigger
DataFrames

```
# add a new column
df ["Job"] = ["Ferrari Driver", "Red Bull Driver", "McLaren Driver"]

# add new rows
new_rows = pd.DataFrame( data: [{"Name": "Lando Norris", "Age": "N/A", "Job": "McLaren Driver"},
                                {"Name": "Lewis Hamilton", "Age": "2000", "Job": "Ferrari Driver"}],
                          index=["Driver 4", "Driver 5"])

df = pd.concat([df, new_rows])

print(df)
```

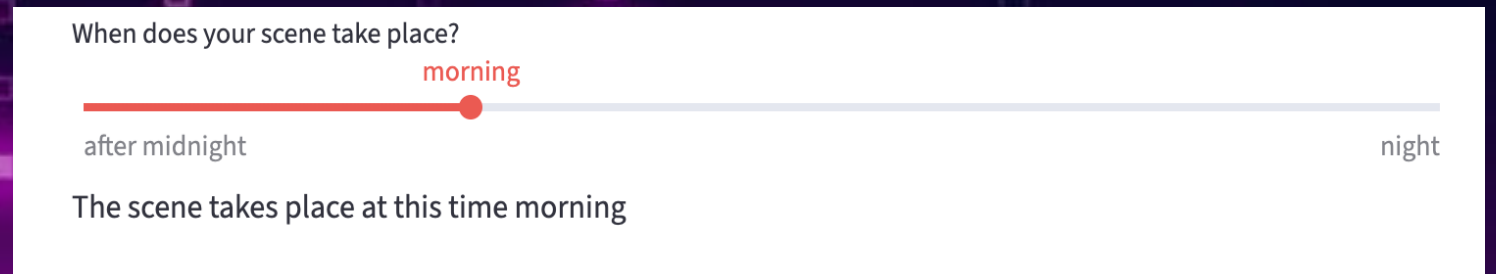
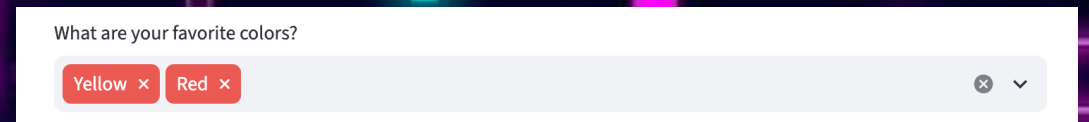
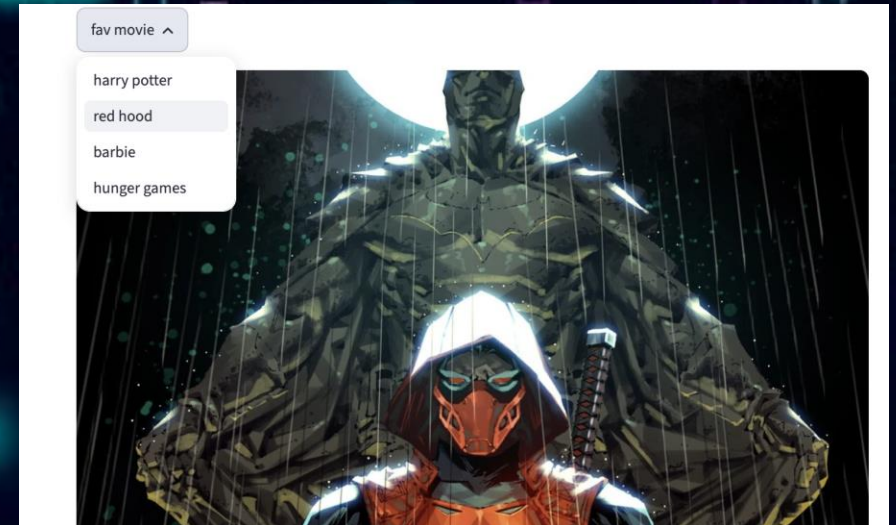
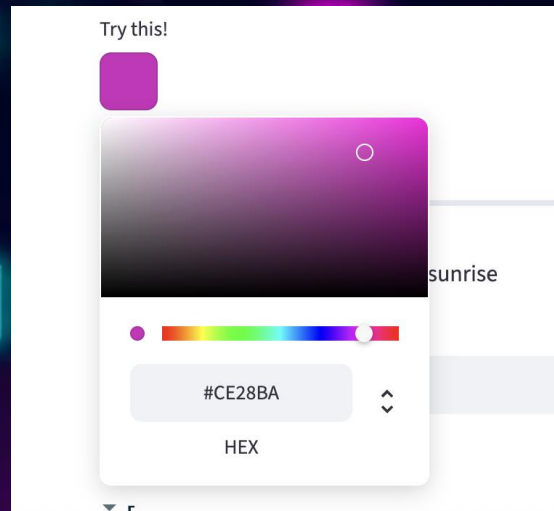
```
df = pd.read_csv( filepath_or_buffer: '/Users /Downloads/Formula1_2025Season_RaceResults.csv', index_col="Driver"
df[["Starting Grid", "Position"]] = df[["Starting Grid", "Position"]].apply(pd.to_numeric, errors="coerce")
poletowin = df[(df["Starting Grid"] < 2) & (df["Position"] < 2)]
print(poletowin.to_string())
```

	Track	Position	No	Team	Starting Grid	Laps	Time/Retired	Points	Set Fastest Lap	Fastest Lap Time
Driver										
Lando Norris	Australia	1.0	4	McLaren Mercedes	1	57	1:42:06.304	25	Yes	1:22.167
Oscar Piastri	China	1.0	81	McLaren Mercedes	1	56	1:30:55.026	25	No	1:35.520
Max Verstappen	Japan	1.0	1	Red Bull Racing Honda RBPT	1	53	1:22:06.983	25	No	1:31.041
Oscar Piastri	Bahrain	1.0	81	McLaren Mercedes	1	57	1:35:39.435	25	Yes	1:35.140
Lando Norris	Monaco	1.0	4	McLaren Mercedes	1	78	1:40:33.843	25	Yes	1:13.221

Coding on a practice website

Messing around with the widget options

```
data = pd.read_csv("/Users/[redacted]/Downloads/mock_shotlist.csv")
st.write(data)
```



Goals Update after I started coding

TO-DO - List

- example shotlist in excel
- figure out how to add data to a csv file when you're using streamlit and pandas
- sidebar navigation
- how to do a pie chart / clock like thing in streamlit / pandas / other library?
- incorporate small HTML details
- st.sessiondata
- figure out how to make all your data into a file that you can download:)
- make it user friendly with the URL concepts!
- this is gonna be cool son
- add a time of the year function
- add the balloon after the download works

Implemented Features: Customizable Project Title

```
x = st.text_input(label="What is the name of your project?", key="hey")
st.write(f"My Project is called {x}")

# reaction and save session

if x:
    st.badge(label="What a cute name!", color="violet")
    st.session_state["x"] = x
    st.success(f"Saved: {x}")
```

```
x = st.session_state.get("x", "")

if x:
    st.title(f" ✨ Shotlist Organizer for {x} / Location ✨ ")
else:
    st.title("Shotlist Organizer / Location")
```

Welcome to your Shotlist Organizer!

What is the name of your project?

Fenia's coding journey

My Project is called Fenia's coding journey

What a cute name!

Saved: Fenia's coding journey

✨ **Shotlist Organizer for Fenia's
coding journey / Location** ✨

Implemented Features:

File Input saving while switching to next page and displaying DF

Paste your csv path here

`/Users/feniachelina/PycharmProjects/TechBasics1_Fenia_Gerhardt/FinalProject/mock_shotlist-2.csv`

Saved:
`/Users/feniachelina/PycharmProjects/TechBasics1_Fenia_Gerhardt/FinalProject/mock_shotlist-2.csv`

Would you rather...?

Sort by Scene! Sort by Location!

Which Location are you looking for?

City Street

Apartment Kitchen Train Station

Ah so this scene is happening there! City Street

	Scene	Scene Description	Location	Shot Number	Shot Size	Category
6	2	Morning Commute	City Street	1	WS (Wide Shot)	EST (Es
7	2	Morning Commute	City Street	2	MCU (Medium Close Up)	Mediur
8	2	Morning Commute	City Street	3	CU (Close Up)	Mediur
9	2	Morning Commute	City Street	4	FS (Full Shot)	Wide S

Implemented Features:

Picture Input for better visual engagement

Paste the URL to your image

No image yet!! Add one above👉

*URL path instead of
file upload on
purpose so the user
can input any
results from the
internet without
needing to download
different pictures
that clutter the
finder*

Paste the URL to your image

com/wordpress/wp-content/uploads/2023/08/chinju-forest-from-genshin-impact.jpg?q=50&fit=crop&w=825&dpr=1.5



Good job!!

Implemented Features: Slider that widges that both does and doesn't access the DataEditor

When does your scene take place? 🌅



The scene takes place at this time sunrise



When does your scene take place? 🌅



The scene takes place at this time golden hour



Which scene are you looking for?



Here is the 2 scene

	≡ Scene	≡ Scene Description	≡ Location	≡ Shot Number	≡ Shot Size	≡ Category
6	2	Morning Commute	City Street	1	MS (Medium Shot)	EST (Es)
7	2	Morning Commute	City Street	2	MCU (Medium Close Up)	Mediur
8	2	Morning Commute	City Street	3	CU (Close Up)	Mediur
9	2	Morning Commute	City Street	4	FS (Full Shot)	Wide S

Which scene are you looking for?



Here is the 7 scene

	≡ Scene	≡ Scene Description	≡ Location	≡ Shot Number	≡ Shot Size	≡ Category
40	7	Forest Search	Forest Clearing	1	WS (Wide Shot)	EST (Es)
41	7	Forest Search	Forest Clearing	2	MCU (Medium Close Up)	Low
42	7	Forest Search	Forest Clearing	3	FS (Full Shot)	Low
43	7	Forest Search	Forest Clearing	4	OTS (Over The Shoulder)	High
44	7	Forest Search	Forest Clearing	5	CU (Close Up)	Close
45	7	Forest Search	Forest Clearing	6	WS (Wide Shot)	High

Implemented Features: Shot List Editor Widgets

Shot Number?

6

You chose Shot Number: 6

Camera Movement?

Static

Gear?

Handheld

Diese Zeile wurde geändert:

	Scene	Scene Description	Location
15	3	Office Confrontation	Corporate O

Shot Size?

MS (Medium Shot)

MS (Medium Shot)

MCU (Medium Close Up)

OTS (Over the shoulder)

WS (Wide Shot)

CU (Close Up)

FS (Full Shot)

Implemented Features: Download Buttons & Add line manually

Add line

Location

Rooftop

Press Enter to apply

Scene

4,00

- +

Shot Number

9,00

- +

Save

Please put in all values

Download as CSV

Download as PDF

saves updated
DataFrame

```
19 def add_line_button(key_suffix):  feniacelina
20     toggle_key = f"show_form_{key_suffix}"
21
22     if toggle_key not in st.session_state:
23         st.session_state[toggle_key] = False
24
25     if st.button(label: "Add line", key=f"toggle_{key_suffix}"):
26         st.session_state[toggle_key] = True
27
28     if st.session_state[toggle_key]:
29         new_location = st.text_input(label: "Location", key=f"location_{key_suffix}")
30         new_scene = st.number_input(label: "Scene", key=f"scene_{key_suffix}")
31         new_shot_number = st.number_input(label: "Shot Number", key=f"shotnumber_{key_suffix}")
32
33         if st.button(label: "Save", key=f"save_{key_suffix}"):
34             if new_location and new_scene:
35                 current_df = st.session_state.get("df")
36                 new_row = pd.DataFrame([{"Location": new_location.strip(),
37                                         "Scene": new_scene,
38                                         "Shot Number": new_shot_number
39                                         }])
39                 st.session_state["df"] = pd.concat(objs: [current_df, new_row], ignore_index=True)
40                 st.session_state[toggle_key] = False
41                 st.rerun()
42
43     else:
44         st.warning("Please put in all values")
45
```

Implemented Features:

What if there are no lines in the file yet?

Your shotlist is empty! Add a few lines

Add line

```
df = st.session_state.get("df", None)
filtering = None

if df is not None and not df.empty:
    location_options = df["Location"].unique().tolist()

    sortlocation = st.select_slider(label="Which Location are you looking for?", options=location_options)
    st.write(*args: "Ah so this scene is happening there!", sortlocation)
    filtering = df[df["Location"] == sortlocation]
    st.data_editor(filtering)
    st.session_state["location"] = sortlocation
    add_line_button(key_suffix="normal_list")
else:
    st.info("Your shotlist is empty! Add a few lines")
    save_line = add_line_button(key_suffix="empty_shotlist")
    st.session_state["save_line"] = save_line
```

Final General Framework

- *The Homepage (main.py): Lets the User name the project, input for the file and lets them decide if they want to sort their shotlist by scene or location*
- *Location: Lets the user display the DataEditor according to the Location Widget*
- *Scene: Lets the user display the DataEditor according to Scene Widget*

Final General Framework

- *Shared Functions: Backend File that stores the functions of both pages that are exactly the same*
- *Location & Scene:*
 - *Picture Display*
 - *DataEditor*
 - *Daytime Display Widget*
 - *Shot Number Selector*
 - *Add Line*
 - *Edit Columns Widgets*
 - *Display updated row*
 - *Download Button*

Difficulties & Problem Solving

- *Keeping track of the logic, especially after changing things to correct structural errors*
- *Developing a clear vision for what I was envisioning and what was doable*
- *Understanding at the beginning 'difficult' streamlit features to the point that I could reuse it*
- *Any widget that had to update the DataEditor*
- *Breaking the issues down step by step*
- *Practicing streamlit features (even ones I didn't use in the end)*
- *Instead of adding every single feature I wanted in the beginning, I decided to select a few and make sure that those work smoothly without many or any errors*

Future Developments

- *Improvement of Design for Website and final PDF file*
- *Add a section for notetaking*
- *Add a feature in which Python calculates the estimated shoot time needed for each scene*

References

- *<https://docs.streamlit.io/develop/api-reference>*
- *External Libraries:*
streamlit, pandas, fpdf, os, sys